

Assignment: Agentic Research Paper Evaluator

1. Problem Statement

The academic community and independent researchers are often overwhelmed by the sheer volume of papers published on repositories like arXiv. Manually verifying the **internal consistency, factual accuracy, and novelty** of a paper is time-consuming.

Existing LLM tools often suffer from limited context windows or "hallucination" when summarizing complex technical documents. There is a need for an **Agentic AI Platform** that doesn't just summarize, but critically audits a research paper by breaking it down into specialized domains of evaluation.

2. Objective

To develop a multi-agent system that autonomously scrapes an arXiv link and executes a comprehensive peer-review simulation. The system must move beyond simple extraction and provide a "Judgement Report" based on cross-referencing internal data and external facts.

3. Scope and Expectations

You are expected to leverage **free-tier LLM providers** (e.g., Gemini Free API, OpenRouter, or local models via Ollama/LM Studio) to power the agents.

Special Consideration: Do not send more than 16k tokens in a single LLM call.

The Platform must execute the following workflow:

- **Web Scraping:** Extracting clean text/data from the provided arXiv URL.
- **Decomposition:** Breaking the paper into sections (Abstract, Methodology, Results, Conclusion).
- **Agentic Analysis:** Utilizing specialized agents for:
 - **Consistency:** Checking if the methodology actually supports the claimed results.
 - **Grammar & Language:** Evaluating professional tone and syntax.
 - **Novelty:** Searching for existing literature to determine uniqueness.
 - **Fact-Checking:** Verifying cited constants, formulas, or historical data.
 - **Authenticity (Accuracy Score):** Calculating a "Fabrication Probability" based on statistical anomalies or logical leaps.

4. Deliverables

A. Codebase

- **Source Code:** A clean, modular repository (GitHub/GitLab).
- **Configuration:** A `README.md` explaining how to set up environment variables for the chosen LLM provider.

- **Framework:** Use of an agentic framework (e.g., CrewAI, AutoGen, or LangGraph) is highly encouraged.

B. Evaluation Report

A PDF or Markdown file generated by the AI for a "Test Case" paper, containing:

- **Executive Summary:** A pass/fail recommendation.
- **Detailed Scores:**
 - **Consistency Score:** (0-100)
 - **Grammar Rating:** (High/Medium/Low)
 - **Novelty Index:** Qualitative description.
 - **Fact Check Log:** List of verified vs. unverified claims.
 - **Accuracy/Fabrication Score:** A percentage-based risk assessment.

C. Live Demo

- **Interface:** A simple UI (Streamlit, Gradio, or a CLI tool).
- **Walkthrough:** A short video or live presentation demonstrating the input of an arXiv link and the real-time generation of the Judgement Report.